A-A-54398
30 April 1991

# COMMERCIAL ITEM DESCRIPTION

## CAVITY LINING AND THINNER SET, DENTAL
Nitrocellulose Base

The General Services Administration has authorized the use of the Commercial Item Description as a replacement for Military Specification MIL-C-37989 (DM) which is cancelled.

This commercial item description covers a nitrocellulose based cavity lining and thinner set suitable for use in dental operative procedures.

Salient characteristics:

Material. The cavity lining shall be composed of the following ingredients:

Ethyl Acetate
Diethyl Ether
Nitrocellulose
Ethyl Alcohol
Toluene Sulfonamide

The ethyl acetate, diethyl ether and ethyl alcohol shall be in accordance with the tests, standards and requirements of the latest edition of the American Chemical Society (ACS) specifications, "Reagent Chemicals".

The nitrocellulose and toluene sulfonamide, which are not included in the ACS, "Reagent Chemicals", shall be of high quality and shall be suitable for the purposes intended. The thinner shall be composed of the following ingredients:

Diethyl Ether
Ethyl Acetate

The ethyl acetate and diethyl ether shall be in accordance with the test standards and requirements of the ACS, "Reagent Chemicals".

The thinner shall be capable of being added to the lining to obtain the original consistency without altering the formula.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Description. The cavity lining and thinner set shall be suitable for use in dental operative procedures. The following components shall be present in each unit of issue of the liner and solvent set.

One bottle containing a minimum of 20 ml of cavity lining.
One bottle containing a minimum of 20 ml of thinner.
One set of instructions for dilution with thinner and for use of cavity lining.

Working properties. The cavity lining shall spread readily over the entire surface of the cavity. It shall adhere to the surfaces and quickly dry to a thin, semipermeable film within 10 seconds of application. It shall not inhibit or alter the polymerization of polymeric restorative material and shall be compatible with dental cavity base materials. The lining shall be capable of sealing the freshly cut dentin surfaces from acid penetration, corrosion products, and leachates so as to protect the dental pulp from toxicity. The lining shall maintain the morphology of the prepared dentin surfaces and be hydrophobic when dry. The lining shall protect exposed surfaces of dentin in teeth that have been prepared for restorations.

Spread. The cavity lining shall have a spread diameter of 13 to 18 millimeters when tested as specified herein.

Percent solids. The cavity lining shall have a percent solids of 7.9 to 8.2% when tested as specified herein.

Viscosity. The cavity lining shall have a viscosity of 3 to 6 centipoises when tested as specified herein.

Instructions. Detailed and accurate instructions for use of lining and thinner shall be furnished.

Certification. A statement shall be forwarded to the procuring activity, prior to award of contract or purchase order, that the lining has been tested clinically and found effective in lining a cavity as described above, and that the lining is free from toxicity and irritation. Supporting data shall be forwarded with the statement or made available on request of the procuring activity. If the data furnished is found inadequate, additional information shall be furnished on request. The Government reserves the right to examine the data, to consult with the test activity and to determine whether such data is satisfactory and acceptable.



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Workmanship. The cavity lining and thinner shall be clean and free from defects which may detract from their appearance or may impair their serviceability.

Unit. Set (SE). One set consisting of one bottle of cavity lining, one bottle of thinner, and instructions for use constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The unit of product for examination purposes shall be one cavity lining and thinner set. Sampling for examination shall be conducted at inspection level II, AQL 2.5.

Examination. Cavity lining and thinner sets shall be examined to determine compliance with specification requirements.



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Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified below. The unit of product for test purposes shall be one cavity lining and thinner set. Sampling for tests shall be conducted at inspection level S-1, AQL 1.0.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Net volume. The net volume of cavity lining and thinner in the bottles shall be measured by a suitable method.

Spread. The spread shall be tested according to the following procedure:

a) Clean a microscope slide with acetone, let dry and place horizontally on a support.

b) Fill an eye dropper to about 1/3 full with the lining, keeping the eye dropper horizontal.

c) Keeping eye dropper 1-1/4 to 1-1/2 inches above the slide, slowly release 1 drop of liquid onto slide. Let it spread and dry.

d) Measure the diameter of the spread.

Percent solids. The percent solids shall be tested according to the following procedure.

a) Weigh a small bottle with lining. Note weight.

b) Weigh an empty aluminum dish.

c) Transfer about 29 grams of lining into the aluminum dish and reweigh the bottle. Note weight. Difference in weight before and after is the exact weight of the lining transfer.

d) Let lining dry in air in the dish for 1-1/2 hours. Reweigh the dish.



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Calculation:

Percent solids =

$$\frac{\text{(Weight of dish with dried lining - weight of empty dish) x 100}}{\text{weight of lining}}$$

Viscosity. The viscosity of the cavity lining shall be tested in accordance with ASTM Designation D301.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.



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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

| National Stock Number | Item Identification |
|---|---|
| 6520-01-192-9472 | CAVITY LINING AND THINNER SET, DENTAL, Nitrocellulose Base |

Ordering data. Intermediate/exterior package quantities, labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA - DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6520-2475
Location: PL#1/CAVITY



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | A-A-54398 | 4-30-91 |

3. DOCUMENT TITLE

CAVITY LINING AND THINNER SET, DENTAL, Nitrocellulose Base

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION |
|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) |
| | e. DATE SUBMITTED (YYMMDD) |

7. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code)<br>(1) Commercial | (2) AUTOVON |
|---|---|---|
| Defense Personnel Support Center | (215)737-2870 | 444-2870 |
| c. ADDRESS (Include Zip Code)<br>ATTN: DPSC-MSTE<br>2800 S. 20th Street<br>Philadelphia, PA 19101 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |

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